

```
import pandas as pd
```

```
df = pd.read_excel("evcarsdataset.xlsx")
```

```
df.head()
```

	Date	Vehicle_ID	Manufacturer	Model	Driver_Name	Vehicle_Status	Revenue_Generated	Maintenance_Cost	Overspeed_Violation
0	14-09-2025	EV031	BYD	Atto 3	Arvind Patel	Charging	0.00	0.0	No
1	23-08-2025	EV085	Hyundai	Kona Electric	Priya Verma	Running	2367.11	0.0	Yes
2	14-08-2025	EV100	BYD	Atto 3	Kavya Das	Running	106.69	0.0	No
3	12-10-2025	EV008	Mahindra	XUV400	Sanjay Patel	Running	2459.14	0.0	Yes
4	08-04-2025	EV036	Hyundai	Kona Electric	Madhavi Rao	Running	655.49	0.0	Yes

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# 1. Running Vehicles Count
running_count = (df['Vehicle_Status'] == 'Running').sum()
print("1. Running Vehicles Count :", running_count)
```

1. Running Vehicles Count : 6013

```
#2.garage vehicles count
garage_count = (df['Vehicle_Status'] == 'Garage').sum()
print("2. Garage Vehicles Count :", garage_count)
```

2. Garage Vehicles Count : 1472

```
#3.Charging and Running Vehicles
charging_count = (df['Vehicle_Status'] == 'Charging').sum()
print("3. Charging Vehicles :", charging_count)
print("    Running Vehicles :", running_count)
```

3. Charging Vehicles : 1501
Running Vehicles : 6013

```
# 4. Manufacturers
print("4. Manufacturers")
print(df['Manufacturer'].unique())
```

4. Manufacturers
['BYD' 'Hyundai' 'Mahindra' 'MG' 'Tata']

```
# 5. Manufacturer Wise Count
print("5. Manufacturer Wise Count")
print(df['Manufacturer'].value_counts())
```

5. Manufacturer Wise Count
Manufacturer
Hyundai 2283
Mahindra 2124
Tata 2070
BYD 2035
MG 1488
Name: count, dtype: int64

```
# 6. Month-wise Revenue Generation by Each Car
df['Month'] = pd.to_datetime(df['Date'], format='%d-%m-%Y').dt.month_name()
print("\n6. Month-wise Revenue")
month_revenue = df.groupby(['Vehicle_ID', 'Month'])['Revenue_Generated'].sum()
print(month_revenue)
```

```
6. Month-wise Revenue
Vehicle_ID  Month      Revenue_Generated
EV001      April      12580.33
           August      6705.56
           December     5862.57
           February     12260.14
           January      7574.66
           ...
EV100      March      11678.68
           May         12523.25
           November     3357.15
           October     13711.86
           September     6351.58
Name: Revenue_Generated, Length: 1200, dtype: float64
```

```
# 7. Month-wise Maintenance
print("\n7. Monthwise Maintenance")
month_maintenance = df.groupby(['Vehicle_ID', 'Month'])['Maintenance_Cost'].sum()
print(month_maintenance)
```

```
7. Monthwise Maintenance
Vehicle_ID  Month      Maintenance_Cost
EV001      April      3515.44
           August        0.00
           December        0.00
           February     3447.65
           January      567.06
           ...
EV100      March      4012.44
           May         0.00
           November     4832.69
           October     2129.09
           September     4250.76
Name: Maintenance_Cost, Length: 1200, dtype: float64
```

```
# 8. Driver with Maximum Revenue
driver_revenue = df.groupby('Driver_Name')['Revenue_Generated'].sum()
print("\n8. Driver with Maximum Revenue")
print(driver_revenue.idxmax())
print("Revenue :", driver_revenue.max())
```

```
8. Driver with Maximum Revenue
Priya Verma
Revenue : 280921.89
```

```
# 9. Driver with Minimum Revenue
print("9. Driver with Minimum Revenue")
print(driver_revenue.idxmin())
print("Revenue :", driver_revenue.min())
```

```
9. Driver with Minimum Revenue
Gokul Patel
Revenue : 165916.64
```

```
df['Overspeed_Violation'] = df['Overspeed_Violation'].map({
    'Yes': 1,
    'No': 0
})
```

```
#10. Driver with Maximum Overspeed Violations
# along with Overspeed Time
overspeed = df.groupby('Driver_Name')['Overspeed_Violation'].sum()
driver = overspeed.idxmax()
print("\n10. Driver with Maximum Overspeed Violations")
print("Driver :", driver)
print("Violations :", overspeed.max())
print("\nOverspeed Times")
```

```
times = df[(df['Driver_Name']==driver) & (df['Overspeed_Violation']>0)][['Date','Time','Overspeed_Violation']]
print(times)
```

10. Driver with Maximum Overspeed Violations
 Driver : Rahul Das
 Violations : 171

Overspeed Times

	Date	Time	Overspeed_Violation
187	07-08-2025	21:10	1
195	11-07-2025	00:26	1
232	03-06-2025	06:53	1
271	10-03-2025	06:18	1
296	14-04-2025	07:55	1
...	...	...	...
9723	10-11-2025	09:22	1
9803	16-03-2025	17:36	1
9840	22-08-2025	11:57	1
9852	18-12-2025	16:10	1
9972	13-10-2025	12:57	1

[171 rows x 3 columns]

```
# 11. DDMMYY Column
df['DDMMYY'] = pd.to_datetime(df['Date'], dayfirst=True).dt.strftime('%d%m%y')
print("11. DDMMYY")
print(df[['Date', 'DDMMYY']].head())
```

11. DDMMYY

	Date	DDMMYY
0	14-09-2025	140925
1	23-08-2025	230825
2	14-08-2025	140825
3	12-10-2025	121025
4	08-04-2025	080425

```
# 12. Present Battery Percentage
print("12. Present Battery Percentage")
print(df[['Vehicle_ID', 'Present_Battery_%']])
```

12. Present Battery Percentage

	Vehicle_ID	Present_Battery_%
0	EV031	17
1	EV085	42
2	EV100	33
3	EV008	33
4	EV036	39
...	...	...
9995	EV043	29
9996	EV014	13
9997	EV077	34
9998	EV035	36
9999	EV083	31

[10000 rows x 2 columns]

```
# 13. Expected Travelling Distance
print("13. Expected Travelling Distance")
print(df[['Vehicle_ID', 'Expected_Travel_Distance_km']])
```

13. Expected Travelling Distance

	Vehicle_ID	Expected_Travel_Distance_km
0	EV031	69
1	EV085	185
2	EV100	125
3	EV008	129
4	EV036	156
...	...	...
9995	EV043	113
9996	EV014	51
9997	EV077	136
9998	EV035	161
9999	EV083	130

[10000 rows x 2 columns]

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(8, 3))
ax.set_xlim(0, 10)
ax.set_ylim(0, 5)
ax.axis('off')
box = plt.Rectangle((1, 0.8), 8, 3.2, fill=False, linewidth=2)
ax.add_patch(box)
ax.text(5, 3.1, "Padma Sree M",
        fontsize=18, fontweight='bold', ha='center')
ax.text(5, 2.0, "Register Number: 727624BAM029",
        fontsize=13, ha='center')

plt.show()
```

